

ME5412 Robotics for Healthcare

Mobility aids

移动辅具

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Outline of the Lecture

- Mobility and mobility limitation
 - Wheelchair basics and classification
 - Manual wheelchairs
 - Power-assist mechanisms
 - Smart wheelchairs
- 课程大纲
- 移动能力与移动功能障碍
 - 轮椅基础与分类
 - 手动轮椅
 - 助力机构
 - 智能轮椅

Mobility

移动能力对每个人的生活质量至关重要，是完成各类活动领域功能的必要条件：

- **Mobility** is fundamental to each individual's quality of life and is necessary for functioning in each of the performance areas:
 - self-care, 自我照料,
 - work or school, 工作或就学,
 - play or leisure. 娱乐或休闲。
- Mobility is a **foundational** activity for many other skills and activities
 - It enables a person to move to or about a location where the activity will be completed, whether that is within a building or in the community and beyond.
- In addition to the functional gain of increased independence in mobility, other goals such as positive self-image, social interaction, and health maintenance are achieved.

移动能力是众多其他技能与活动的基础行为

它使人能够前往活动开展场所并在该场所内活动，无论是建筑内部、社区或是更远的地方。

除获得行动独立带来的功能提升外，还可以实现积极自我形象、社会交往、维持健康等其他目标。

个人移动能力

Personal Mobility

个人移动能力指个体自主从一处移动到另一处的能力，可以在建筑内、建筑周边、建筑外部或建筑之间移动。

- **Personal mobility** is the ability to move oneself from one place to another, within a building, around and outside a building, or between buildings. 个人移动能力区别于借助私人或公共交通工具实现的移动。
- Personal mobility is distinguished from mobility achieved through the use of any form of transportation, private or public
- The World Health Organization's International Classification of Functioning, Disability and Health (ICF) identifies mobility as an **activity**, rather than an impairment of a body structure or function (WHO, 2001) 世界卫生组织《国际功能、残疾和健康分类（ICF）》将移动能力界定为一项活动，而不是身体结构或身体功能的损伤（WHO，2001）。

移动功能障碍.

Mobility Limitations

多国使用移动辅具的人群数量显著增长，源于三大趋势：

Significant increase in the number of individuals using mobility aids, across many countries, related to 3 trends:

1. the increasing proportion of older adults in many countries
 - The proportion of individuals reporting a disability increases substantially with age
 - Age-related physical changes such as arthritis result in mobility impairments
2. the rising rates of obesity
 - Mobility devices specifically designed to support the increased size and weight of these individuals
 - **Bariatric chairs** for these individuals whose mobility is impaired by obesity and related chronic diseases
3. accessibility legislation
 - Mandating construction of physical environments that are accessible to individuals who use a wheelchair
 - Defining funding mechanisms to support the acquisition of mobility devices.

1. 多国老年人口占比上升

• 随年龄增长，存在残疾问题的人口占比大幅上升

• 关节炎等年龄相关身体变化会造成移动功能受损

3. 肥胖率上升

• 专门为体型、体重更大的人群设计的移动设备

• 肥胖专用轮椅，面向因肥胖及相关慢性病导致行动受限的人群

3. 无障碍相关立法

• 强制要求建设可供轮椅使用者通行的物理环境

• 设立资金机制，支持人们购置移动辅具。

移动功能障碍的诱因

Causes of Mobility Limitations

- 神经系统疾病 脑卒中、脑瘫、吉兰-巴雷综合征、亨廷顿舞蹈症、脑外伤、肌营养不良、帕金森病、脊髓灰质炎、脊髓损伤、脊柱裂、多发性硬化。
- Neurological disorders
 - Stroke, cerebral palsy, Guillain-Barré syndrome, Huntington's chorea, traumatic brain injury, muscular dystrophy, Parkinson's disease, poliomyelitis, spinal cord injury, spina bifida, and multiple sclerosis
 - Symptoms commonly seen in these neurological disorders are muscle weakness or paralysis, sensory deficits, and abnormal muscle tone 这类神经系统疾病常见症状：肌肉无力或瘫痪、感觉缺损、肌张力异常。
- Musculoskeletal disorders
 - Orthopedic and rheumatological conditions, ankylosing spondylitis, osteogenesis imperfecta, osteoporosis (loss of bone density), Paget's disease, and scoliosis, lower extremity amputation, acquired or congenital
 - Symptoms commonly seen in individuals with arthritis include painful, swollen, and stiff joints (particularly in hand and wrist), muscle weakness, muscle wasting around the affected joints, complaints of feeling fatigued, and, in later stages, joint contractures resulting in range-of-motion limitations
- Cognitive disorders
 - Concomitant disorders that affect an individual's cognitive functioning and ability to learn, such as Alzheimer's disease and cognitive impairment
- Chronic disorders like diabetes, cardiorespiratory conditions, and obesity
 - Fatigue or restrictions related to energy expenditure
 - Amputations resulting from complications due to diabetes

Bear in mind that not all individuals with a given diagnosis experience a similar impairment in mobility

Causes of Mobility Limitations

- 肌肉骨骼系统疾病

骨科与风湿类病症：强直性脊柱炎、成骨不全症、骨质疏松（骨密度下降）、佩吉特骨病、脊柱侧凸、后天或先天性下肢截肢。

关节炎患者常见症状：关节疼痛、肿胀、僵硬（尤其手与腕关节）；肌肉无力、病变关节周围肌肉萎缩；易疲劳；晚期出现关节挛缩，造成关节活动范围受限。

- 认知障碍

合并病症，会影响人的认知功能与学习能力，例如阿尔茨海默病以及认知功能损伤。

- 慢性病，例如糖尿病、心肺疾病、肥胖

- 能量消耗带来的疲劳或行动受限
- 糖尿病并发症引发的截肢。

需要注意：即便确诊同一疾病，不同患者的移动功能受损程度并不相同。

移动辅具

Mobility Aids

功能移动能力受限可以通过两种方式应对：

- Limitations to functional mobility can be
 - **augmented** with low-tech AT such as canes, walkers, or crutches
 - 使用低技术辅助器具进行能力增强，例如手杖、助行器、腋拐
 - **replaced** by wheeled mobility systems of various types
 - 使用各类轮式移动系统替代原有移动能力
- Lack of access to mobility devices may be a limitation to full participation in society
 - 无法获得移动辅具，会阻碍人们充分参与社会生活。



Mobility aids



- Limitations to functional mobility can be
 - **augmented** with low-tech AT such as canes, walkers, or crutches
- 机器人技术可以借助智能电动轮椅系统，提升人的移动能力。
- **Robotics technologies** can contribute to enhance an individual's mobility with smart power wheelchair systems.

Scope of Mobility Limitations

完全可步行者：无移动功能障碍

- **Full ambulator:** no mobility impairment

边缘步行者：

- **Marginal ambulator:**

- can walk short distances
- may need wheelchair at times, particularly outside the home

- 可以短距离行走
- 部分场景需要轮椅，尤其居家以外环境

- **Manual wheelchair user:**
- has some method of propelling a manual wheelchair, whether it is with (1) both upper extremities, (2) both lower extremities, or (3) one upper and one lower extremity , or a caregiver

• 手动轮椅使用者：

• 具备驱动手动轮椅的方式：可以使用①双侧上肢、②双侧下肢、③一侧上肢+一侧下肢，或依靠照护人员推行。

- **Marginal manual wheelchair user:**

- manual wheelchair is used part of the time and powered wheelchair part of the time

混合型手动轮椅使用者：

- 部分时间使用手动轮椅，部分时间使用电动轮椅

- **Totally/severely mobility-impaired user:**

- unable to propel self independently in a manual wheelchair
- dependent mobility system (propelled by an attendant) or
- Independent powered mobility system

• 完全/重度移动障碍使用者：

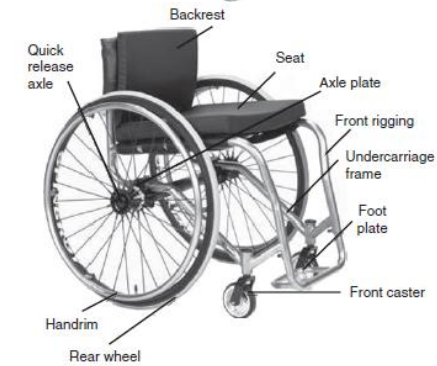
无法独立驱动手动轮椅；

使用照护人员推行的依赖型移动系统，或是独立电动移动系统。

Mobility Aid Classification

Three broad categories of wheeled mobility systems:

- **Dependent mobility systems** are propelled by an attendant and include strollers and transport chairs, as well as manual wheelchairs
A dependent mobility system is chosen when (1) the individual is not at all capable of independently propelling a wheelchair or (2) a secondary system is needed that is lightweight and easily transported.
- An **independent manual mobility system** is for those individuals who have the ability to propel a wheelchair manually.
These bases have two smaller front wheels and two larger wheels in the back (drive wheels) that allow the user to propel independently.
- **Independent power mobility systems** are required when the user has difficulty propelling a manual wheelchair.
These are **power wheelchairs** that are driven by the user.



Mobility Aid Classification

轮式移动系统分为三大类：

- 依赖型移动系统：由陪护人员推行；包含手推婴儿车、转运椅，以及手动轮椅。
有人推，自己用不了，无电动

选择依赖型移动系统的两种情形：①使用者完全无法独立驱动轮椅；②需要一套轻便、便于转运的备用系统。

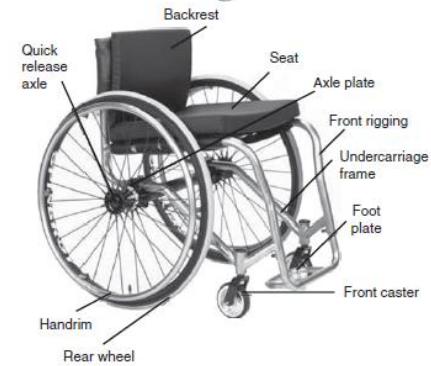
- 独立手动移动系统面向有能力手动驱动轮椅的使用者。

该类轮椅底盘拥有两个较小前轮，后方两个大驱动轮，可供使用者自主驱动。
无人推，自己可以用

- 独立电动移动系统：当使用者难以驱动手动轮椅时使用。

即由使用者自主操控的电动轮椅。

自用 + 电动



轮椅主要部件

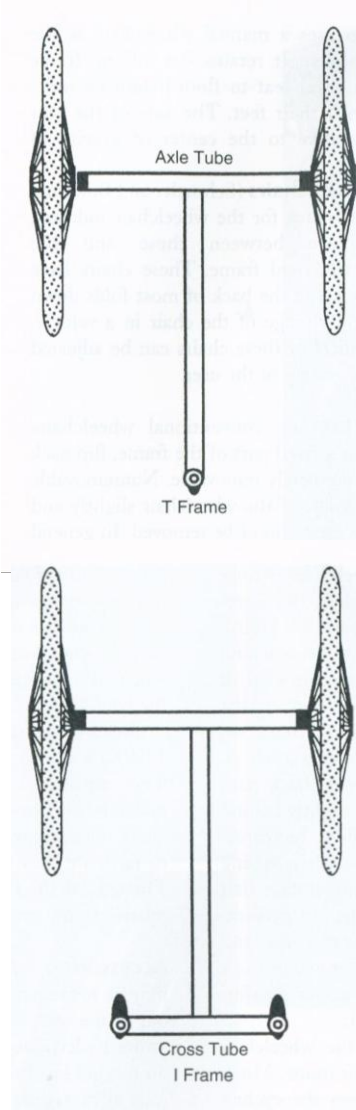
Wheelchair Main Components

Supporting structure

支撑结构.

支撑结构 Supporting Structure

- 车架：
 - 车架类型：刚性车架 / 可折叠车架
 - 后轮轴位置：可调节 / 不可调节
 - 轮椅车架材料：钢材、铝合金、其他材料
 - 车架样式：箱型、悬臂式、T型、I型车架
- 附属配件（扶手、脚踏板等）
- Frame:
 - type of frame: rigid or folding
 - position of the axle of the rear wheels: adjustable or not
 - material used to construct the wheelchair frame: steel, aluminum, other
 - frame style: box, cantilever, T or I frame
- Attachments (armrests, footrests, ...)
- (Optional) Specialized seating and positioning
 - (可选) 专用坐姿与体位支撑组件



轮椅部件标注

Wheelchair Anatomy

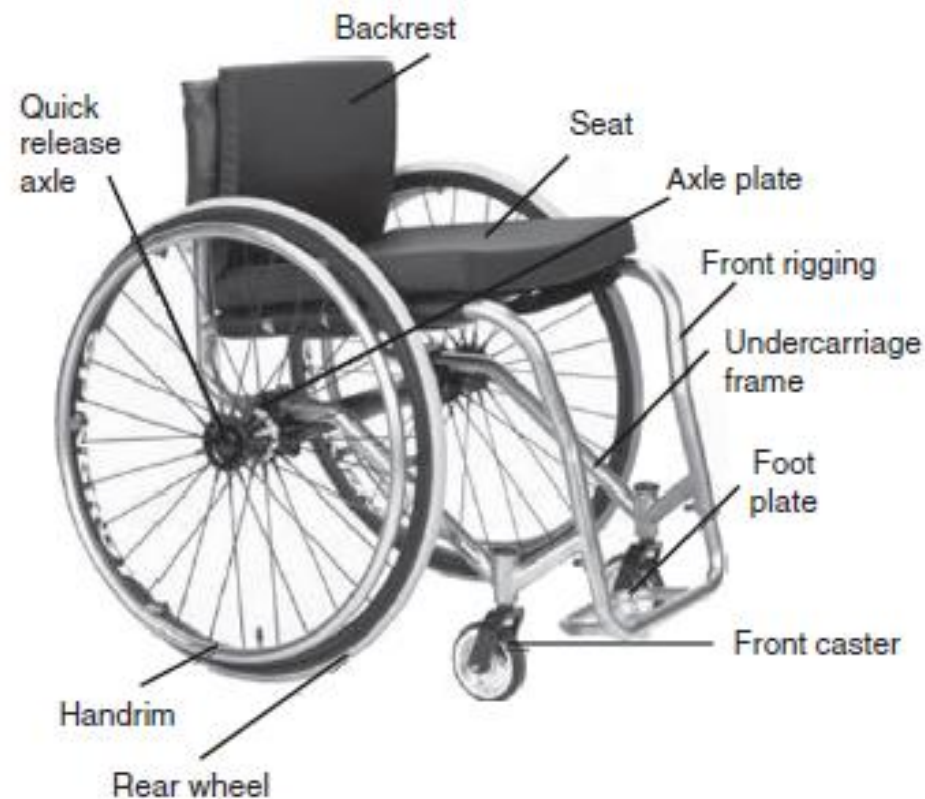
Folding manual wheelchair

可折叠手动



刚性手动

Rigid frame manual wheelchair



可倾斜、可后仰支撑结构

Tilting and Reclinable Supporting Structures

A. with **tilt** feature

Ability to rotate a specific seating position around a fixed axis

A. 具备整体倾斜功能

座椅整体绕固定转轴旋转，保持坐姿形态。



B. with **recline** feature

Changing the seat-to-back angle, resulting in greater than 90 degrees

B. 具备靠背后仰功能

改变座椅与靠背夹角，夹角可大于90°。



轮椅分类.

Wheelchair Classification

- Standard wheelchairs
 - Folding steel chairs
 - Fixed axle of rear wheels
 - Heavy and not useful for long-term use
 - Short-term use, e.g., in airports, malls, ...
- Lightweight wheelchairs
 - Steel/aluminium alloy frames
- Ultra-lightweight wheelchairs
 - Axle of rear wheels adjustable to the user's centre of gravity
 - Useful for an individual who uses a manual wheelchair as the primary means of mobility
- Rigid sport ultra-lightweight wheelchairs
 - Rigid frame
 - Titanium/carbon fiber composites
 - Axle of rear wheels adjustable to the user's centre of gravity

- 标准轮椅
钢制折叠轮椅
后轮轴不可调节
重量大，不适合长期使用
短期使用，例如机场、商场等场所

- 轻量化轮椅
钢/铝合金车架

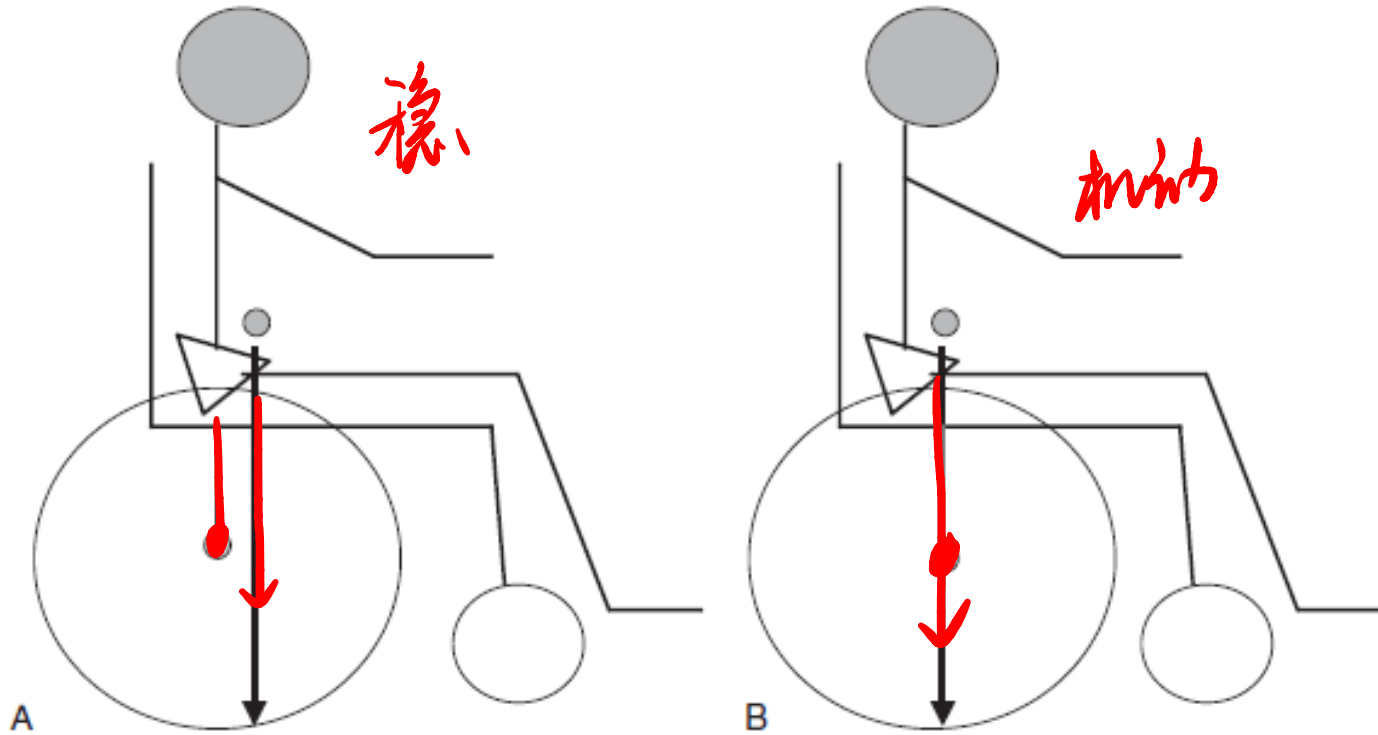
- 超轻量轮椅
后轮轴位置可根据使用者重心调节
适合将手动轮椅作为主要移动方式的使用者

竞技型刚性超轻量轮椅

- 刚性车架
- 钛合金/碳纤维复合材料
- 后轮轴可匹配使用者重心进行调节

使用者重心与后轮轴的位置关系

Relationship of the User's Centre of Gravity to the Axle of Rear Wheels



A. 使用者坐姿下重心位于轮椅车轴前方，轮椅稳定性更高。

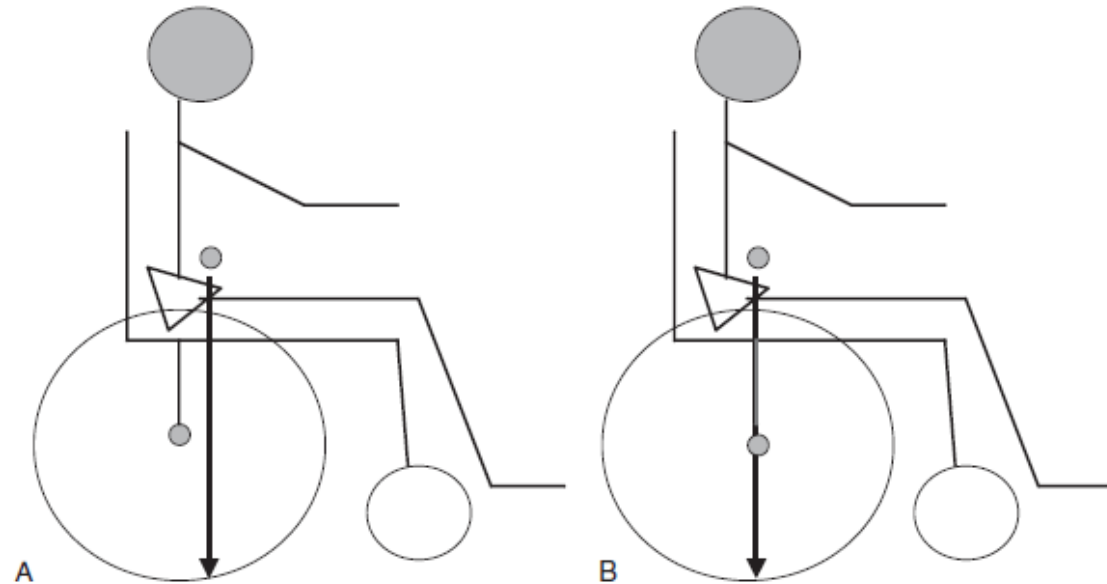
A. When the user is seated with his center of gravity ahead of the axle of the wheelchair, the chair is **more stable**.

B. When the user is seated with his center of gravity directly above the axle of the wheelchair, the chair is **more maneuverable**.

B. 使用者坐姿下重心正好处于轮椅车轴正上方，轮椅机动性更好。

Relationship of the User's Centre of Gravity to the Axle of Rear Wheels

A or B?



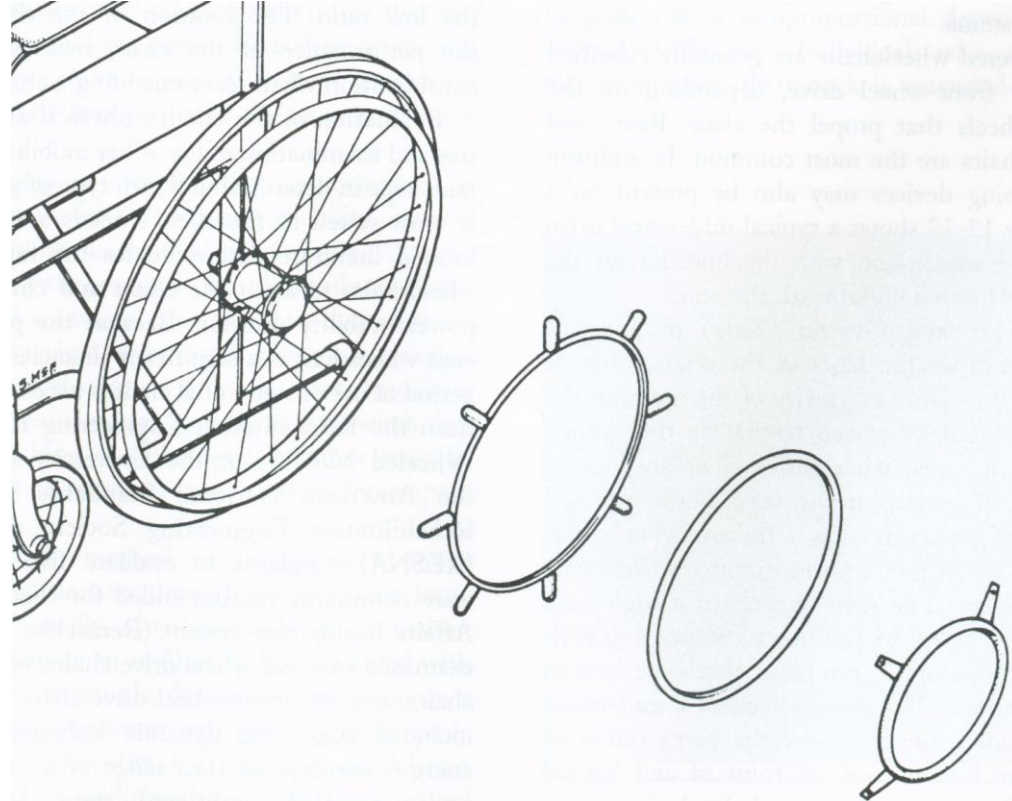
Wheelchair Main Components

Propelling structure

驱动结构

Propelling Structure of Manual Wheelchairs

Hand rims 手推圈



助力机构

Power-assist Mechanism

- Manual wheelchair use may cause shoulder injuries
 - 使用手动轮椅可能造成肩部损伤
- Power-assist wheels amplify the user's pushing force
 - 助力轮可以放大使用者的推力。



助力机构公式

Power-assist Mechanism

$$\tau_m(t) = k_m i(t)$$

$\tau_m(t)$: motor torque

k_m : motor torque constant 转矩系数

$i(t)$: motor current 电机电流

$$\text{output torque} = k_g \tau_m(t)$$

k_g : gear ratio 传动比

$$u = k_w k_g \tau_m(t) = k_w k_g k_m i(t)$$

u : control signal 控制信号

k_w : wheel ratio

车轮传动系数

$$i(t) \xrightarrow{k_m} \tau_m(t) \xrightarrow{k_g} \text{gear output} \xrightarrow{k_w} u$$

Power-assist Mechanism – Hand Rim Input

$$u = k_w k_p \tau_p(t)$$

u : control signal

k_w : wheel ratio

k_p : power-assist ratio 助力放大系数

$\tau_p(t)$: hand rim torque 手推圈输入转矩

$$u = 0 \quad \left| \tau_p(t) \right| < \varepsilon \quad \text{当手推圈转矩绝对值小于阈值}\varepsilon, \text{控制输出}u=0$$

$$u = k_w k_p \tau_p(t) \quad \left| \tau_p(t) \right| \geq \varepsilon \quad \text{当手推圈转矩绝对值大于等于阈值}\varepsilon, \text{输出}$$

$$k_w k_g k_m i(t) = k_w k_g \tau_m(t) = u = k_w k_p \tau_p(t)$$

★ 电动轮椅.

Power Wheelchairs

- 车轮由电机驱动
 - Power drive to the wheels
 - Power is delivered to one pair of wheels (i.e., the drive wheels) with additional pairs of wheels providing stability
 - Classified as:
 - Rear-wheel drive
 - Mid-wheel drive
 - Front-wheel drive
- 动力输出至其中一对车轮（驱动轮），其余车轮负责保障稳定性
 - 分类：
 - 后轮驱动
 - 中轮驱动
 - 前轮驱动



电动轮椅控制接口

Power Wheelchair Control Interfaces

- 360°方向控制：操纵杆拨向哪个方向，轮椅就朝该方向运动。

①

- 使用连续操纵杆实现比例控制

- 操纵杆偏离中位的幅度越大，轮椅行驶速度越快。

- **Proportional control with a continuous joystick**

- 360-degree directionality, which means that the chair moves in whichever direction the continuous joystick is displaced.

- The greater the joystick displacement with respect to its rest position, the faster the chair moves

②

- 使用离散操纵杆或开关实现非比例控制
- 同一时刻仅能激活一个运动方向

- **Nonproportional control with a discrete joystick or switches**

- Movement is activated one direction at a time

- 用户想要改变方向时，必须松开原有方向控制，再激活新方向。

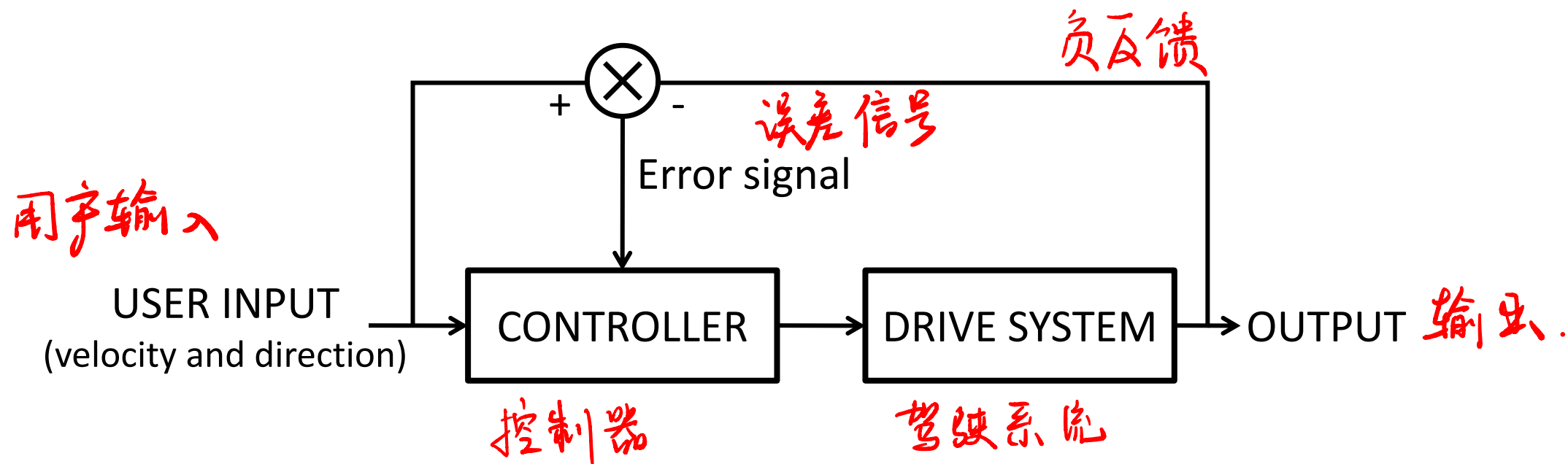
- If the user wishes to change direction, he/she must release the control in one direction and activate it in the direction of the change

- This type of control is typically used when a user needs to drive the chair using switches.

- 该控制方式多用于使用者依靠开关操控轮椅的场景。

电动轮椅控制

Power Wheelchair Control



How to make a power wheelchair robotic?

0

A standard power wheelchair to which a computer and a collection of sensors have been added

方案1: 在标准电动轮椅上加装计算机与各类传感器

0%

A mobile robot base to which a seat has been attached

方案2: 在移动机器人底盘上加装座椅

0%

Both of the above possible



0%

None of the above: newly designed system

0%

★ 智能轮椅

Smart Wheelchairs

- 两种实现方案：
 - 在标准电动轮椅上加装计算机与各类传感器
 - 在移动机器人底盘上加装座椅
- Two approaches:
 - a standard power wheelchair to which a computer and a collection of sensors have been added
 - a mobile robot base to which a seat has been attached
- Useful for wheelchair users who have
 - low vision or severely restricted visual field,
 - 低视力或视野严重受限,
 - motor impairments such as excessive tone or tremor, or
 - 运动功能障碍, 例如肌张力过高、肢体震颤,
 - cognitive impairments that limit the ability to navigate a wheelchair safely
 - 认知障碍, 无法安全操控轮椅完成导航。

☆ 两大功能

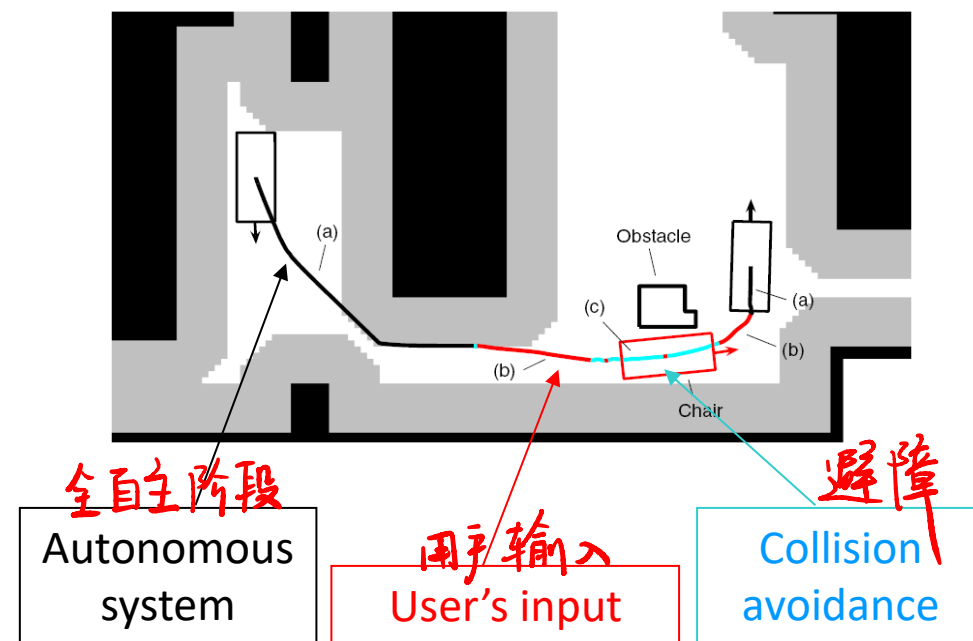
Smart Wheelchairs

- Two functions:
 - **collision avoidance**: sensors detect an obstacle in the path of the wheelchair and will slow or stop the chair or provide some form of warning if the user does not perform a maneuver to avoid it
 - **navigation**, either along a path or through obstacles: the sensors guide the chair along a pathway that is programmed into the chair's control system or that follows an environmental path, or the sensors guide the chair through a feature such as a doorway.

避障：传感器检测轮椅行进路径上的障碍物；若使用者没有做出避让操作，轮椅会减速、停止，或者发出警告。

导航（沿路径行驶或穿越障碍物）：传感器引导轮椅沿着控制系统预设路径、或是环境里的参考路径行驶；也可引导轮椅通过门口这类环境结构。

Shared - control.



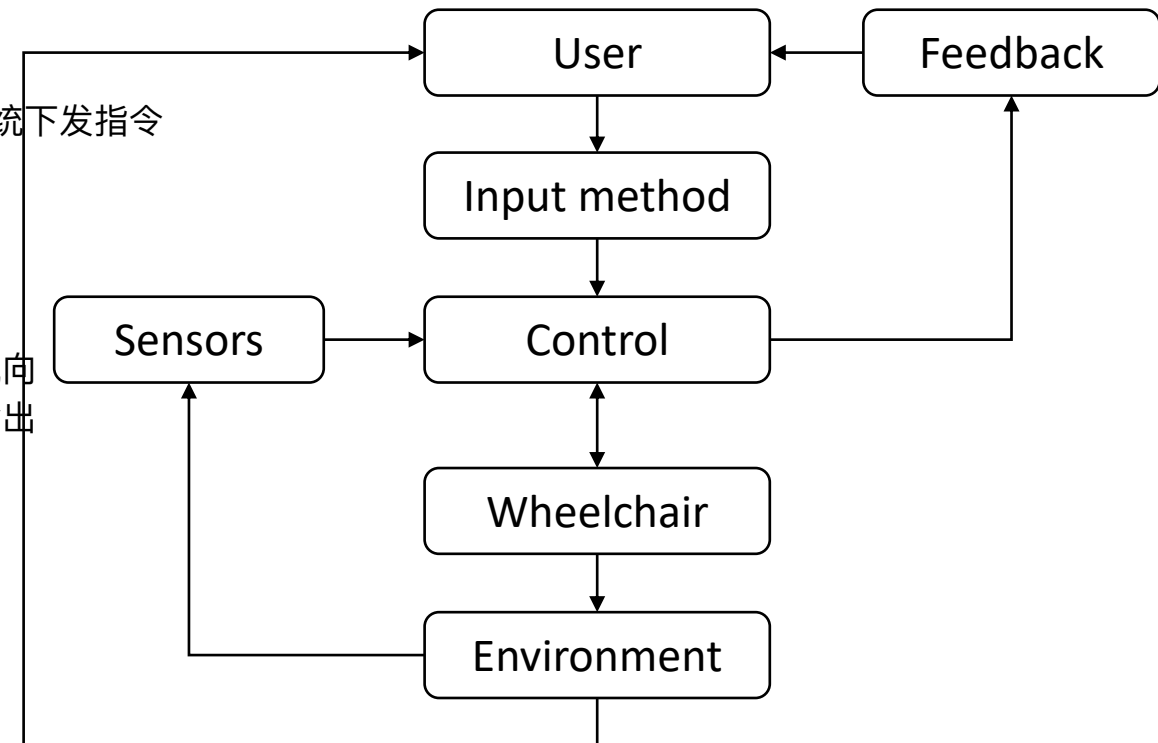
智能轮椅的组成

Smart Wheelchair Components

- 传感器：识别环境特征
- **Sensors** for identifying features in the environment
- **Input methods** for the user to provide commands to the system
- Methods for the system to provide **feedback** to the user
- **Control** algorithm that coordinates inputs from the user, the sensors and the PWC to determine the system behaviour

• 输入方式：使用者向系统下发指令

• 反馈机制：系统向使用者输出反馈



• 控制算法：协调来自用户、传感器、电动轮椅本体的输入，决定系统行为。

☆ 智能轮椅的传感器

Smart Wheelchair Sensors

A variety of sensors have been used in smart wheelchairs, but no sensor (or combination of sensors) has been identified that is sufficiently accurate, inexpensive, small, lightweight, and impervious to environmental conditions (e.g., lighting, precipitation, temperature) and requires little power.

TABLE 5.1 Summary of Sensors Used by Smart Wheelchairs

Sensor	Capability	Advantages	Disadvantages
Ultrasonic (sonar)	Measures distance to obstacles by emitting sound waves	• Small • Low power • Low cost	• Sensitive to obstacle properties (such as sound-absorbing material) • Cannot detect very small/thin or concave obstacles • Cross-talk issues with other sounds or multiple ultrasonic sensors • Cannot detect drop-offs unless sensor is pointed to the floor • Limited coverage
Infrared (IR)	Measures distance to obstacles by emitting light; can also detect transitions from light to dark lines	• Small • Low power • Low cost	• False positives in natural light (due to IR interference) • Sensitive to flooring materials (issues with dark, light-absorbent, transparent, and refractive surfaces) • Limited coverage
Laser range finder	Provides 180° scan of obstacles (distances) in the environment	• High precision • Good for longer ranges	• High cost • High power • Possible eye safety issues • Only detects objects at the height of the laser range finder
Stereovision camera	Captures color or black-and-white image and depth images of the environment	• Low power • Medium cost • Can be used for high-level scene understanding (e.g., object and location recognition) and head/eye tracking of users	• Cannot detect textureless objects • Cannot perform in poorly lit conditions • Challenges posed by reflective and transparent surfaces

超声波

红外

雷达

立体视觉

智能轮椅已经使用过多种传感器，但目前没有任何一种传感器（或传感器组合）可以同时做到精度足够、价格低廉、体积小、重量轻、功耗低，并且不受光照、降水、温度等环境条件影响。

Smart Wheelchair Sensors (contd.)

TABLE 5.1 Summary of Sensors Used by Smart Wheelchairs

Sensor	Capability	Advantages	Disadvantages
Structured light (e.g., Microsoft Kinect)	Captures color or black-and-white image and depth images (using projected IR) of the environment	• Low power • Low cost • Can be used for high-level scene understanding (e.g., object and location recognition) and head/eye tracking of users	• Challenges posed by outdoor environments (due to IR interference) • Sees through transparent surfaces (i.e., cannot be used to avoid collisions with glass doors) • Can demand a lot of processing power
Bump (tactile sensor)	Can detect if contact is made (force is exerted) by an object	• Low cost • Low power	• Requires contact (safety issues) • Increasing coverage (e.g., by using bumper skirt-like sensors) can lead to increased form factor and bulkiness
Wheelchair encoders	Measures number and speed of wheel rotations	• Can be used to easily gather information on distances traveled and speed	• Can be expensive to retrofit on wheelchairs • Wheel slippage can lead to errors
Inertial measurement units (IMUs)	Contain accelerometers, gyroscopes, and optional magnetometers that can be used to determine velocity, orientation, and position, respectively	• Low cost • Low power	• Prone to accumulated errors when used for longer distances • Higher-accuracy IMUs can be expensive
Global positioning system (GPS)	Provides location and time information	• Can give accurate location when GPS satellites are in an unobstructed line of sight • Low cost	• Does not work well indoors

结构光
传感器
(深度相机)

碰撞传感器
不是 Force Sensor

编码器

IMU

GPS

智能轮椅的输入方式

智能轮椅已经研究多种替代方案，摆脱电动轮椅传统输入方式（操纵杆、气动开关）。

Smart wheelchairs have been used to explore a variety of alternatives to the more “traditional” input methods associated with power wheelchairs (e.g., joystick, pneumatic switches).

Smart Wheelchair Input Methods

TABLE 5.2 Alternative Input Methods Used in Smart Wheelchairs

Input	Capability	Advantages	Disadvantages
Brain-computer interface 脑机接口	Sensors may be worn on the head (e.g., electroencephalography, EEG) or implanted in the skull (e.g., electrocorticography, ECoG). 脑电信号 (脑皮层电图 ECoG)	Requires no physical movement	算力高. - Requires significant computational power - Expensive - Limited signals available for control - Sensors must be placed on or inside the skull
Eye gaze 眼动相机	Cameras pointed at the user's face track where the user is looking to determine where the PWC should steer. 追踪视线方向.	- Requires limited physical movement - No sensors touch the user's body	高算力 - Requires significant computational power - Distinguishing between “looking to drive” and “looking to see” is challenging - Expensive 贵.
Voice recognition 语音识别	Speech recognition software interprets steering commands from user.	- Low cost - No sensors touch the user's body	说话清晰. - Requires ability to speak clearly and consistently - Voice has limited bandwidth

智能轮椅反馈形式.

Smart Wheelchair Feedback Modalities

• 智能轮椅可以通过多种模式、多种交互接口向使用者反馈信息：

- Smart wheelchairs can provide feedback to the driver through various modalities and interfaces:

扬声器输出音频反馈。

- **speakers** can be used to provide audio feedback

LED灯实现视觉反馈，例如距离障碍物过近时红灯闪烁。

- small light-emitting diodes (**LEDs**) can be used to offer visual feedback (such as a flashing red light if the driver gets too close to an object)

• 触摸屏：既接收使用者输入（例如目标地点），也展示信息（日程、寻路辅助、虚拟环境等）。

- **haptic** feedback can be provided by using small vibrations to warn the driver about obstacles

• 触觉反馈：微小振动，提醒使用者前方存在障碍物。

- **touch screen displays** can be used both to gather information from the driver (e.g., desired destination) and to display information (e.g., scheduled activities, way-finding assistance, virtual environments, etc.).

Smart Wheelchair Feedback Modalities (cntd.)

TABLE 5.3 Summary of Feedback Modalities Used by Smart Wheelchairs

Modality	Implementation	Advantages	Disadvantages
听觉 听觉	Feedback can be provided through intermittent sound (e.g., warning sirens), continuous sound (e.g., tone indicating distance to the nearest obstacle), or spoken words. 报警、提示	• Does not distract visual attention • Low cost	• May be washed out by ambient noise • Nonspoken feedback may be misinterpreted • Only available for a short period of time
视觉 视觉	Feedback can be provided through something as simple as individual lights or as complex as a “dashboard” implemented on a computer screen. 屏幕仪表盘	• Not affected by ambient noise • Remains available to user • Low cost	• Can distract visual attention
触觉 触觉	Feedback can be provided through pressure applied <u>to a joystick</u> .	• Possible to give very precise feedback about the environment around the PWC • Does not distract visual attention	• Only works if the driver uses a joystick to operate the PWC • Expensive

智能轮椅控制.

Smart Wheelchair Control

- 根据智能轮椅具备的功能与搭载传感器，存在多种控制软件实现方案。
- A variety of approaches to implement control software for smart wheelchairs, based on the functions supported by the smart wheelchair and the sensors it uses
- Two main control approaches: 两大主流
 - Autonomous control ① 全自动
 - Shared control, or semiautonomous control ② 共享控制 (半自动)

Smart Wheelchair Control

Autonomous control system:

- The onboard computer assumes full control of the chair
- Sensors are used to guide the chair

智能轮椅控制-自主控制系统

- 车载计算机全权接管轮椅控制
- 依靠传感器引导轮椅运动。

Name a technique that can be used for autonomous navigation of a smart wheelchair

✓ 0

Nobody has responded yet.

Hang tight! Responses are coming in.

Smart Wheelchair Control

Autonomous control system:

- The onboard computer assumes full control of the chair
- Sensors are used to guide the chair

Shared control (or semiautonomous) system:

- The control of the wheelchair is shared between the user and the onboard computer
- It consists of combining or switching between input from the driver and the intelligent system
- This method of control offers the driver higher levels of control while ensuring safety
- It may be more acceptable to users because they retain some control and do not experience the sensation of the chair moving without their guidance
- However, the exact mechanism of semiautonomous control that is both effective and acceptable to users is still under investigation

Smart Wheelchair Control

共享控制（半自主）系统：

- The control of the wheelchair is shared between the user and the onboard computer
- 轮椅控制权由使用者与车载计算机共同分担。
- It consists of combining or switching between input from the driver and the intelligent system
- 将使用者输入与智能系统输入进行融合或者切换。
- This method of control offers the driver higher levels of control while ensuring safety
- 该控制模式在保障安全的前提下，让使用者拥有更高操控权限。
- It may be more acceptable to users because they retain some control and do not experience the sensation of the chair moving without their guidance
- 用户接受度更高，使用者保留部分控制权，不会出现轮椅不受自身操控自行移动的感受。
- However, the exact mechanism of semiautonomous control that is both effective and acceptable to users is still under investigation
- 但是，兼顾有效性和用户接受度的半自主控制具体实现机制仍在研究当中。

Possible smart wheelchair shared control



The user drives, the smart wheelchair ensures that obstacles are avoided

The user delegates control to the smart wheelchair when going through doors or along hallways

switch between driver and Intelligent System.

The user drives and the smart wheelchair gives warnings on close obstacles

严格来说不算

The user sets a destination and/or waypoints and the smart wheelchair navigates to them autonomously

感觉像 Autonomous.

The user sets a direction and the smart wheelchair follows the direction avoiding obstacles, until a new direction is set

讨论

Smart Wheelchairs – Discussion

- As might be anticipated, the cost of these systems is extremely high
 - 正如预期，这类系统成本极其高昂。
 - A significant technical obstacle to commercialization is the cost/accuracy trade-off that must be made with existing sensors
 - 商业化一大技术阻碍：现有传感器必须在成本与精度之间做取舍。
 - Available plug and play technology is likely to reduce this cost in the near future
 - 现成即插即用技术有望在不久将来降低成本。
 - There is reason for optimism, however, that the significant research in self-driving cars will lead to affordable technology that can also be used in smart wheelchairs
 - 不过可以保持乐观：自动驾驶汽车领域大量研究，有望产出成本可控的技术，同样可以用于智能轮椅。
- In 2014, the technology used in one of Google's self-driving cars cost approximately \$320,000, but industry reports predicted that the additional price a consumer will pay for self-driving car technology will be approximately \$5,000 by 2030 (IHS Automotive 2014)
- 2014年谷歌自动驾驶汽车整套技术成本约32万美元；行业报告预测到2030年，消费者为自动驾驶技术额外支付的费用大约仅5000美元（IHS汽车，2014）。

Since approximately two-thirds of the cost of the technology underlying self-driving cars is actually devoted to GPS technology, the additional price of technology for a smart wheelchair might be as low as \$2,000

由于自动驾驶汽车近三分之二成本来自GPS相关技术；智能轮椅不需要高性能GPS，其附加技术成本可以低至2000美元。

Smart Wheelchairs – Discussion

- Smart technology for wheelchairs is not available on the commercial market*, existing only in research settings

- 轮椅智能技术尚未大规模商用，大多仅存在于科研环境。

Smart wheelchairs are at TRL4**: low-fidelity systems used to demonstrate basic functionality in controlled environments

智能轮椅技术成熟度处于TRL4（技术就绪等级4级）：在受控环境下，使用低保真系统演示基础功能。

- However, research continues to make this technology more accessible, reliable, and safe.

- 但相关研究仍在持续推进，目标让这项技术更易用、更可靠、更安全。

** validated in a laboratory environment

**TRL4含义：仅在实验室环境完成验证。

Smile 智能系统

Smile Smart System



搭载Smile智能系统（SSS）的电动轮椅，具备循线行驶与防碰撞传感器。

- Power wheelchairs provided with the Smile Smart System (SSS) features, including **line following** capacity and **anti-collision sensors**
- Smile Smart System controls provide a powerchair with safety, as it facilitates autonomous movement regardless of vision levels or muscle patterns.

Smile智能系统控制器保障电动轮椅安全；无论使用者视力水平、肌肉状态如何，都可以辅助实现自主移动。



<https://www.smilemart-tech.com/product/smilesmartssystem-powerchairs/>

Smile Smart System

SMILE
Smart
SYSTEM

Key System Elements



Controls 控制器

Allow numerous input options with simple selection and no need for computer programming.

支持多种输入方式，操作简单选择即可，无需编程。



Bumper 防撞缓冲保险杠

Protects the chair and the driver. Over time the bumper can be removed as driving skills improve.

保护轮椅与使用者；当使用者操控技术提升后，可以拆除保险杠。



Sensors 传感器

Detect objects in front of the powerchair to avoid collisions and reduce damage.

检测轮椅前方物体，规避碰撞，降低设备损坏风险。



Tape 导向胶带

Indoor and outdoor, permanent or temporary guidance system to steer the powerchair and avoid collisions.

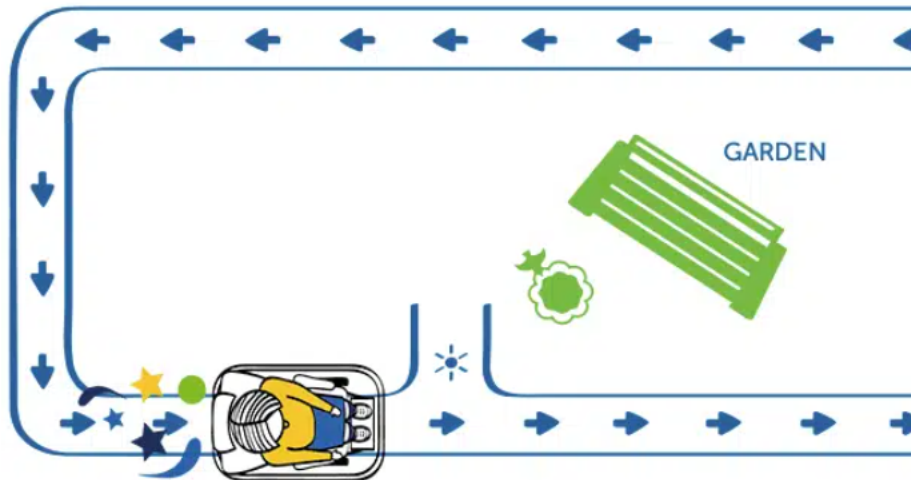
室内室外均可使用；可永久或临时布设，作为引导系统，引导轮椅行驶，规避碰撞。



<https://www.smilemart-tech.com/product/smilesmartsystem-powerchairs/>

Smile Smart System

Line following



Explore



Line following can be used as a safe pathway from which to develop switch access skills and joystick use over time, gently adapting settings as personal abilities evolve.

Smile Smart System

Anti-collision sensors



Motivate



Anti-collision sensors and bumpers are a brilliant way to prevent damage. Fewer repairs means reduced costs, limited interruptions and allows more freedom & smiles.

Smile Smart System



Personal Drive Prescription

Combining all the available elements of the Smile Smart System, our team are able to go that step further by creating unique mounting, switching, controls and accessory formations.

The extensive range of adaptive switches, control devices and mounting solutions at Smile Smart Technology enable users to complete a wide range of daily tasks with ease.

Many of the switches and control solutions have been developed in-house by Smile Smart Technology. Our holistic approach to mounting along with a commitment to good aesthetics help us provide user-friendly solutions, all based on the feedback from users and support teams.

Summary

Questions ?

- Personal mobility is a key activity for ADLs
- 个人移动能力是日常生活活动（ADL）中的核心行为。
- Mobility limitations are growing, to a variety of health conditions, and restrict an individual's participation
- 各类健康问题造成移动功能障碍人群数量持续增长，障碍会限制个体参与社会活动。
- Wheelchairs are the most common mobility aid, replacing (not augmenting) the individual mobility
- 轮椅是最常见的移动辅具，它是替代而非增强人的原有移动能力。
- Wheelchairs range from manual to power wheelchair Robotic technologies can contribute to smart power wheelchair, with obstacle avoidance and navigation capability, with autonomous or shared control
- 轮椅分为手动轮椅与电动轮椅；机器人技术可以赋能智能电动轮椅，实现避障、导航，支持自主控制或者共享控制模式。